Phase 3 (Extended Reach) Subtask Deepening — epic → task → subtask (closes R5)

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Revision: 2 **Last modified:** 2026-07-04T12:00:00Z **Rev 2:** Independent gap-analysis pass — structure and honesty stance (`PENDING_DEVICE:`/`Operator-blocked` vocabulary) verified consistent with `09-phase3-reach-wbs.md` and the sibling Phase-1/Phase-2 deepening docs. No contradictions found.

> Volume 7 (Phase Execution), document 5 of 5. Companion to `subtask-deepening-p1.md` / `-p2.md`, applying the R5 deepening (`REFINEMENT_NOTES.md`) to Phase 3: every task `HVPN-P3-NNN` in `09-phase3-reach-wbs.md` is decomposed into PR-sized subtasks `HVPN-P3-NNN.k` (§11.4.93/.54), each with a concrete **acceptance** (falsifiable, captured-evidence §11.4.5/.69/.107), the **§11.4.169 test types** (the `09-…` §1 vocabulary incl. the Phase-3-specific `REPRO`), and an **estimated complexity** (engineer-day T-shirt, **widest error bars in the programme**, sizing-only `TARGET`, §11.4.6). **Spec-only.** Phase 3 is the one phase where "*ship what's provable, flag the rest honestly*" is the correct stance (`09-…` §13): the two native shims (E21 HarmonyOS, E22 Aurora) are real native work on toolchains the project does not yet own hardware/CI for, so device-gated acceptances are marked **`PENDING_DEVICE:`** (§11.4.3 `hardware_not_present`) and engagement-gated ones **`Operator-blocked`** (§11.4.21/.101) — never a faked pass. Every subtask traces to its parent task in `09-…`. These `.k` rows feed the `workable-items` DB (`workable-items-model.md` §7).

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0. Deepening conventions + the honesty stance

Columns as in `-p1`/`-p2` §0: `id` (`HVPN-P3-NNN.k`) · **Subtask** (≥6 words §11.4.91) · **Acceptance** · **Tests** (§11.4.169 codes, `09-…` §1 vocab incl. `REPRO`) · **Cx** (XS/S/M/L engineer-days, sizing `TARGET`).

Phase-3 honesty stance (§11.4.6, `09-…` §0/§13). Where a subtask's acceptance genuinely cannot be proven without the target device or an external engagement, the acceptance carries the literal `PENDING_DEVICE:` (device-gated, §11.4.3 `hardware_not_present`) or the item's status is `Operator-blocked` (engagement / spend, §11.4.21/.101) with the §11.4.148 D3 unblock condition. A gate is **never** marked `pass` on metadata/config-only evidence; the runtime signature (§11.4.108) for the native shims is asserted on a **clean install on the real device** — an APK/HAP/RPM that *builds* is not *done*. No active CI (§11.4.156): the "fork runners" are local/self-hosted rootless-Podman build pods, never GitHub/GitLab workflows.

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1. E20 — Reach CI fabric (gate G20)

Self-hosted rootless Podman build pods (§11.4.76/.161/.156); a fork lag never blocks mainline.

HVPN-P3-200 — Pin OpenHarmony-SIG Flutter fork + DevEco toolchain in a pod (`09-…` §5; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	`harmonyos.container` quadlet building a HAP	`podman build` produces a HAP from the sample	UNIT,FA	M

id	Subtask	Acceptance	Tests	Cx
	from the pinned `flutter_flutter ohos` SHA	app; toolchain SHA recorded		
`.2`	DevEco signing material mounted read-only as secrets (§11.4.10)	secrets never in the image layer (SEC leak audit green)	SEC	S
`.3`	REPRO: same SHA → same toolchain digest	identical toolchain digest on rebuild (REPRO)	REPRO	S

HVPN-P3-201 — Pin OMP Aurora `flutter-aurora` fork + RPM signing pod (`09-…` §5; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	`aurora.container` quadlet producing a signed Aurora RPM	signed RPM of the sample app builds; `rpm -K` verifies the signature	UNIT,FA	M
`.2`	Mirror-pin the Russian-hosted SDK (§11.4.77 regen) + isolated egress allow-list	SDK tarball SHA-256 matches the recorded manifest; egress allow-list holds	SEC,REPRO	S

HVPN-P3-202 — Reach release-artifact registry + provenance manifest (`09-…` §5; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	Signed `reach_artifacts.json` (source SHA + toolchain digest + SBOM pointer per HAP/RPM/.wasm)	manifest entry per build, schema-validated, links resolve	UNIT,FA	S

HVPN-P3-203 — G20 readiness certification (`09-…` §5; XS)

id	Subtask	Acceptance	Tests	Cx
`.1`	`make reach-ci-gate` builds+signs both fork artefacts, emits the G20 verdict	HAP + signed RPM both produced from a clean checkout in two `-count=2` runs; gate `outcome=pass`	FA,CHAL	XS

2. E21 — HarmonyOS NEXT build (G21, highest risk)

ArkTS VPN ability + NAPI bridge to the Rust `helix-core` `.so`; the UI ports for free, the tunnel ability is real native work.

HVPN-P3-210 — `helix-core` cross-compile to OpenHarmony + NAPI surface (`09-…` §6; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	Build `helix-ffi` for `aarch64-unknown-linux-ohos`; emit `libhelix_ohos.so` NAPI module (re-exports the platform-neutral core, §11.4.28)	`.so` loads in an ArkTS test harness; `helix_version()` round-trips	UNIT,INT	M
`.2`	`helix_start(fd,cfg)`/ `helix_stop`/ `helix_subscribe` NAPI surface over the unchanged core	NAPI surface callable from ArkTS (INT)	UNIT,INT	M
`.3`	MEM/BENCH on emulator	RSS ceiling sampled (`PENDING_DEVICE:` on hardware); throughput benched on emulator	MEM,BENCH	M

HVPN-P3-211 — Network Kit `VpnExtensionAbility` + lifecycle + kill-switch (`09-…` §6; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	ArkTS ability: VPN consent → `VpnConnection` (tun fd/ routes/DNS, MTU 1280) → hand fd to core	ability creates the tunnel fd + drives the core (INT)	UNIT,INT	M
`.2`	Drive connect/ disconnect/ kill-switch/ auto-	`PENDING_DEVICE:` enroll→UP→`curl` reaches authorized LAN host on a HarmonyOS NEXT	E2E,MEM,UX,REC	L

id	Subtask	Acceptance	Tests	Cx
	escalate from the core status stream	device; window-scoped MP4 + DevEco heap capture; §11.4.3 SKIP `hardware_not_present` until G20 provisions a device		
`.3`	Tunnel-drop blanks plaintext (kill-switch per §04)	`PENDING_DEVICE`: kill-switch blanks plaintext on drop (device capture)	SEC,CHAL	M

HVPN-P3-212 — Flutter HAP build (Access + Connector, OHOS channel) (`09-…` §6; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Signed HAPs via `runHelixApp(flavor,…)` on the SIG fork; ArkTS↔Dart MethodChannel wired	app installs + drives the shim; golden UI tests pass on the OHOS theme	UI	M
`.2`	UX walkthrough MP4 (`PENDING_DEVICE`)	UX MP4 vision-verified (§11.4.159), `PENDING_DEVICE`: until a device	UX,REC	S

HVPN-P3-213 — DevEco signing + market metadata + l10n (zh-Hans first-tier) (`09-…` §6; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	DevEco signing profile + AppGallery metadata + `intl_zh.arb` first-tier locale	signed HAP verifies; zh-Hans strings render with no overflow (UI golden)	FA,UI	S

HVPN-P3-214 — G21 device certification + honest gap doc (`09-…` §6; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	`challenges/helix_qa` bank: enroll→UP→reach→kill-switch on device, PASS only on captured	gate `outcome=pass` with device evidence, **OR** `outcome=pending_device` with the §11.4.148 D3 unblock condition	E2E,FA,CHAL,REC	M

id	Subtask	Acceptance	Tests	Cx
	evidence (§11.4.27/.107)	(provision a HarmonyOS NEXT device + add to the §11.4.128 tracked-device set) — never a faked pass		

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3. E22 — Aurora OS build (G22)

OMP Russia Flutter fork → signed RPM; tunnel backend in Qt/C++ linking `helix-core` as a C library. Enterprise/government SKU.

HVPN-P3-220 — `helix-core` C ABI for Aurora + cbindgen header (`09-…` §7; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	Build `helix-ffi` for the `aarch64`/ `armv7hl` Sailfish target; `cbindgen` → `helix.h` (no Aurora specifics inside the core, §11.4.28)	header compiles in a Qt test; `helix_version()` round-trips	UNIT,INT	M
`.2`	MEM/BENCH on the Aurora target	RSS sampled (`PENDING_DEVICE:`); throughput benched	MEM,BENCH	M

HVPN-P3-221 — Qt/C++ `tun` lifecycle + Friflex Flutter bridge + kill-switch (`09-…` §7; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	Open/ route/DNS the Sailfish `tun`; `helix_start` over the C ABI; status marshalled onto the Qt event loop	the backend drives the core; status events surface (INT)	UNIT,INT	M
`.2`	Friflex-style plugin	`PENDING_DEVICE:` enroll→UP→reach on an Aurora	E2E,MEM,UX,REC,CHAL	L

id	Subtask	Acceptance	Tests	Cx
	bridge connect/ disconnect/ status to Flutter + kill-switch/ auto-escalate	device; kill-switch blanks plaintext; MP4 + RSS capture; §11.4.3 SKIP until a device		

HVPN-P3-222 — Flutter signed-RPM build (Access + Connector, OMP fork) (`09-…` §7; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Signed Aurora RPMs of the two flavors + `ru` first-tier l10n	`rpm -K` verifies; `ru` strings render with no overflow; install on the Aurora emulator (`PENDING_DEVICE`: for hardware reach)	UI,UX,REC,SEC	M

HVPN-P3-223 — G22 device certification + enterprise-SKU ops doc (`09-…` §7; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Bank entry enroll→UP→reach→kill-switch on Aurora + ops doc (isolated runner + Mos.Hub signing §11.4.10)	gate `outcome=pass` with device evidence, **OR** honest `pending_device`; toolchain-provenance recorded `UNCONFIRMED:` until an Aurora device is in hand (§11.4.6)	E2E,FA,CHAL,REC	M

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4. E23 — WASM browser-scoped MASQUE proxy (G23)

Browser-scoped proxy of the browser's own traffic — NOT a system-wide tunnel; the reuse pillar: `helix-transport` → wasm32 over WebTransport.

HVPN-P3-230 — `helix-transport` wasm32 build + WebTransport binding (`09-…` §8; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	Feature-gate `helix-transport` to compile to `wasm32-unknown-unknown`; bind browser WebTransport for the H3/datagram path (same MASQUE state machine, only the I/O leaf differs)	`.wasm` loads in headless Chromium; a MASQUE `CONNECT-UDP` session establishes to a test edge	UNIT,INT	L
`.2`	Throughput benched	proxy MB/s captured (BENCH)	BENCH	S

HVPN-P3-231 — Service-worker / fetch-shim integration + scope isolation (`09-…` §8; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Service worker routing *only* explicitly-proxied origins through MASQUE; everything else normal browser networking	a non-proxied origin never touches the gateway (SEC, captured)	UNIT,INT,E2E	M
`.2`	Hard scope: only policy-authorized hosts (server-side need-to-know); die on tab/worker close	a non-authorized host is unreachable; tab-close terminates the session, no lingering tunnel (SEC)	SEC	M

HVPN-P3-232 — Console UX: explicit "browser-scoped, not a system VPN" affordance (`09-…` §8; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	`ShieldIndicator` variant + copy stating the limitation plainly; per-network toggle; live reachability	UX MP4 vision-verified that the scope wording is present + unambiguous (§11.4.159) —	UI,UX,REC	S

id	Subtask	Acceptance	Tests	Cx
		guards an over-claim bluff		

HVPN-P3-233 — G23 certification + threat note (WASM origin model) (`09-…` §8; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	Bank: browser reaches an authorized host, denied an unauthorized one, leaks nothing beyond scope; threat note (short-lived token, no WG private key in browser)	gate `outcome=pass` with captured evidence; the "no key in browser" invariant proven by a SEC assertion	E2E,SEC,CHAL	S

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5. E24 — Billing-optional multi-tenant (G24)

Billing must not break no-logging — metering is aggregate counters only; billing OFF by default = zero behaviour change.

HVPN-P3-240 — Billing schema (plans/subs/entitlements/aggregate meters, RLS + no-log lint) (`09-…` §9; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	FORCE-RLS tenant-scoped billing tables; `usage_counter` daily-aggregate only (no src/dst/host/port/flow, no sub-day ts)	cross-tenant read denied (RLS); aggregate schema passes the lint	UNIT,INT,SEC	M
`.2`	Extend the no-log CI lint to reject a billing column	the lint FAILs a planted `usage_flows(src,dst,port,ts)` table, passes the aggregate schema	SEC	S

id	Subtask	Acceptance	Tests	Cx
	matching the traffic-log shape + paired §1.1 mutation			

HVPN-P3-241 — Aggregate metering pipeline (edge counters → daily upsert) (`09-…` §9; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Metering consumer folds edge rx/tx counters into `usage_counter` daily upserts (idempotent by tenant/device/day)	exactly-once bucket under a SIGKILL mid-fold (CHAOS)	UNIT,INT,CHAOS	M
`.2`	Bounded-memory at scale	10k devices, bounded memory (STRESS)	STRESS	S

HVPN-P3-242 — Entitlement enforcement in the coordinator (`09-…` §9; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Plan limits → admission checks at enroll + map build; over-quota = deny with reason (D-P3-2 fail-open)	an over-device-quota enroll is denied with a clear reason; bytes-quota breach degrades per the chosen policy	UNIT,INT,SEC	M
`.2`	PERF: check adds < 1 ms to map build	captured map-build delta < 1 ms	PERF	S

HVPN-P3-243 — Optional payment-provider adapter (interface + one reference impl, flagged) (`09-…` §9; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Narrow `PaymentProvider` interface + one reference adapter (provider behind a	billing-OFF path never calls the adapter (captured)	UNIT,INT,SEC	M

id	Subtask	Acceptance	Tests	Cx
	flag, no name in core §11.4.28)			
.2	Public webhook: signature-verify + replay-reject + rate-limit (first public surface → DDOS applicable)	webhook signature verified; replay rejected; rate-limit holds under flood (DDOS)	DDOS,SEC	S

HVPN-P3-244 — G24 certification (billing ON works, OFF is a perfect no-op, no-log preserved) (`09-…` §9; S)

id	Subtask	Acceptance	Tests	Cx
.1	Bank: (a) billing-ON meter+quota+payment flow; (b) billing-OFF byte-identical to pre-E24 (§11.4.108 runtime-signature diff); (c) no-log lint still green + planted traffic-log still FAILs	gate `outcome=pass`; the no-op proof captured as a §11.4.135 regression guard	E2E,SEC,FA,CHAL	S

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6. E25 — Third-party security audit (G25)

A self-audit is not an audit; the verdict is the external party's. Engagement is operator-gated (spend → §11.4.101/.66).

HVPN-P3-250 — Audit scope, threat model, asset inventory (`09-…` §10; M)

id	Subtask	Acceptance	Tests	Cx
.1	STRIDE per surface (crypto core, control plane, the four reach surfaces, no-log invariant)	every shipped surface mapped to ≥1 threat + ≥1 audit objective; no surface unlisted (§11.4.118)	SEC	M

HVPN-P3-251 — SBOM + reproducible-source snapshot for the auditor (`09-…` §10; S)

id	Subtask	Acceptance	Tests	Cx
.1			SEC,REPRO	S

id	Subtask	Acceptance	Tests	Cx
	CycloneDX SBOM per artefact + a pinned rebuildable source snapshot	SBOM covers 100% of shipped deps; snapshot SHA recorded; `endor`/`aikido`-class scan green or every finding triaged		

HVPN-P3-252 — Engage auditor + run the engagement (operator-gated) (`09-…` §10; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	Select + contract an independent firm (operator decision, §11.4.66 `AskUserQuestion`), share scope+snapshot, receive the report	report received; status `Operator-blocked` with the §11.4.21 unblock detail until the engagement is funded — honest, not faked	SEC,CHAL	L

HVPN-P3-253 — Remediation loop to zero Critical/High (§11.4.134/.146) (`09-…` §10; L)

id	Subtask	Acceptance	Tests	Cx
`.1`	Per finding: §11.4.102 root cause → §11.4.115 RED-on-vulnerable → fix → §11.4.146 extend → permanent §11.4.135 guard	zero unresolved Critical/High; each closure carries a falsifiable regression test (no console-mark-is-fixed bluff)	UNIT,INT,SEC,CHAL,REC	L
`.2`	Re-audit fixed items until the firm signs off (§11.4.134 iterate-to-GO)	the firm signs off after re-audit	SEC,CHAL	M

HVPN-P3-254 — G25 certification + public disclosure (`09-…` §10; S)

id	Subtask	Acceptance	Tests	Cx
.1	Publish the report + remediation summary; gate verdict	gate `outcome=pass` only when Critical/High = 0 + report published; else honest `fail`/`operator_blocked` with the open-finding list	SEC,CHAL	S

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7. E26 — Reproducible builds (G26)

Table stakes for a privacy product: a user must verify the binaries match the source.

HVPN-P3-260 — Determinize Go + Rust builds (`09-…` §11; M)

id	Subtask	Acceptance	Tests	Cx
.1	Pin toolchains, `-trimpath`/`--remap-path-prefix`, fix `SOURCE_DATE_EPOCH`, strip non-determinism, build in a pinned `containers` image	two clean rebuilds of `helix-go`+`helix-edge`+`helix-core` on two hosts → identical SHA-256 (REPRO)	REPRO,FA	M

HVPN-P3-261 — Determinize the Flutter app builds (`09-…` §11; M)

id	Subtask	Acceptance	Tests	Cx
.1	Reproducible Flutter AOT (iOS/Android/desktop/web) + fork HAP/RPM; deterministic asset + ARB ordering	two rebuilds of each flavor → identical artefact (signature stripped before compare); divergences root-caused (§11.4.102) to zero	REPRO,FA	M
.2	Honest boundary: signed installers differ by signature → compare the unsigned payload digest	the honest boundary recorded (§11.4.6); unsigned-payload digests identical	REPRO	S

HVPN-P3-262 — WASM + native-shim reproducibility (`09-…` §11; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	Pin OHOS/Aurora cross-toolchains; rebuild `.wasm` + `libhelix_ohos.so` + Aurora `libhelix.a`	two rebuilds → identical SHA-256, **OR** `PENDING_TOOLCHAIN`: honestly where a fork toolchain is not yet deterministic (a finding, not a pass)	REPRO	M

HVPN-P3-263 — Public verification script + transparency doc (`09-…` §11; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	`verify-reproducible.sh` + `reproducible_builds.md` (model + honest signature exceptions)	a third party (the §11.4.165 independent verifier) runs it + confirms identity on every non-signature artefact (captured)	REPRO,FA,CHAL	S

HVPN-P3-264 — G26 certification (`09-…` §11; XS)

id	Subtask	Acceptance	Tests	Cx
`.1`	Gate verdict over the full artefact matrix	gate `outcome=pass` only when every non-signature artefact reproduces bit-identically on two independent hosts; any `PENDING_TOOLCHAIN` keeps the gate honestly open	REPRO,FA,CHAL	XS

8. E27 — Reach l10n, governance, release

HVPN-P3-270 — First-tier `ru` + `zh-Hans` l10n completeness + overflow audit (`09-…` §12; M)

id	Subtask	Acceptance	Tests	Cx
`.1`			UI,UX,REC	M

id	Subtask	Acceptance	Tests	Cx
	Complete `helix_l10n` ARBs for `ru`/`zh-Hans`; no overflow/overlap (§11.4.162); state announced not just colored (a11y, §04)	golden UI green in both locales on Access/Connector/Console; UX MP4 vision-verified		

HVPN-P3-271 — Phase-3 feature Status + Status_Summary set (§11.4.153) with per-surface video confirmation (`09-…` §12; M)

id	Subtask	Acceptance	Tests	Cx
`.1`	`docs/features/Status.md` + `Status_Summary.md` enumerating every Phase-3 surface (Implementation/Wiring/Validation/Video-confirmation), four-format export (HTML+PDF+DOCX), docs-chain-synced	every confirmed row has a real-use MP4 (`PENDING_DEVICE` rows honestly marked); §11.4.86 fingerprint matches	REC,CHAL	M

HVPN-P3-272 — docs-chain sync + items-DB load + reproducible-release tag (`09-…` §12; S)

id	Subtask	Acceptance	Tests	Cx
`.1`	Load all `HVPN-P3-*` rows into `docs/workable_items.db` (§11.4.93/.95), sync exports (§11.4.65), cut `<prefix>-3.0.0-…` (§11.4.151) + fan out via merge-onto-latest-main (§11.4.113, never force-push)	`workable-items validate` green; tag prefix correct across main + every owned submodule; artefacts reproduce (E26)	FA,REPRO	S

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9. Subtask roll-up + honest-gap ledger

Epic	Parent tasks	Deepened subtasks	Honest-gap class
E20 reach CI fabric	4	7	toolchain access (medium)
	5	11	

Epic	Parent tasks	Deepened subtasks	Honest-gap class
E21 HarmonyOS NEXT			**`PENDING_DEVICE` — make-or-break**
E22 Aurora OS	4	6	**`PENDING_DEVICE` + Russian toolchain `UNCONFIRMED`**
E23 WASM proxy	4	6	browser API maturity (medium)
E24 billing-optional	5	9	no-log invariant must hold
E25 third-party audit	5	7	**`Operator-blocked` — external spend**
E26 reproducible builds	5	7	fork-toolchain `PENDING_TOOLCHAIN`
E27 reach l10n/release	3	3	low
Total	**35**	**56**	—

Honest-gap ledger (§11.4.6, `09-…` §13/§14 risk register). Phase 3 is the phase where some acceptances cannot be self-proven; the deepening preserves that honesty in the subtask acceptances rather than hiding it:

Subtask	Gap	Honest marker	Unblock (§11.4.148 D3)
`211.2`/`.3`, `214.1` HarmonyOS device reach	no HarmonyOS NEXT device/CI	`PENDING_DEVICE:` / gate `pending_device`	provision a device + add to the §11.4.128 tracked set (via G20)
`221.2`, `223.1` Aurora device reach	no Aurora device + Russian toolchain	`PENDING_DEVICE:` + `UNCONFIRMED:` provenance	provision an Aurora device; mirror-pin + audit the SDK
`252.1` audit engagement	external spend	`Operator-blocked` (§11.4.101/.66)	fund + contract an independent audit firm
	fork toolchain not		root-cause the fork

Subtask	Gap	Honest marker	Unblock (§11.4.148 D3)
`262.1`, `264.1` fork-toolchain reproducibility	yet deterministic	`PENDING_TOOLCHAIN:` keeps G26 honestly open	non-determinism to zero

None of these may be marked `pass` on metadata/config-only evidence (`09-…` §13). A device-/engagement-gated gate that cannot be proven is `pending_device`/ `operator_blocked` with its unblock condition — the release notes state plainly which reach targets are *certified* vs *built-but-pending-device*. Complexity sums to ≈ `09-…` §14's ~172 engineer-days — the widest error bars in the programme, sizing only, never a date (§11.4.6).

Sources verified

- `09-phase3-reach-wbs.md` §2.1 gates G20–G26, §5–§12 (every task Desc/Deliverable/Acceptance/Effort/Tests), §13 traceability + honest-gap rule, §14 risk register, §15 open decisions D-P3-* — read 2026-06-26.
- Sibling `workable-items-model.md` (§3/§6/§9 incl. `gates` table + `pending_device`/`operator_blocked` outcomes), `dependency-graph.md` (§6 Phase-3 DAG), `subtask-deepening-p1.md`/`-p2.md` (§0 conventions) — authored this volume.
- Constitution anchors §11.4.3/.6/.10/.21/.27/.28/.40/.54/.66/.76/.86/.91/.93/.101/.102/.107/.108/.113/.115/.118/.128/.134/.135/.146/.151/.153/.156/.159/.161/.162/.165/.169 — read 2026-06-26.

> Honest boundary (§11.4.6): subtasks decompose stated Phase-3 work. Device- and engagement-gated subtasks carry `PENDING_DEVICE:` / `UNCONFIRMED:` / `Operator-blocked` markers verbatim — a gate is never `pass` on metadata-only evidence. All complexity is sizing `TARGET`, never a date; Phase-3 carries the programme's widest error bars.